Finite Groups and Their Representations — Exercise Sheet 13

Symmetric functions and character rules

Variant: Questions only

Exercise 1 (The tables of S_4 and S_5 by coefficient extraction). Use the Frobenius formula of Lecture 13 to reconstruct the complete tables of S_4 and S_5. Order rows and columns by decreasing lexicographic order of partitions. An exact symbolic computation is permitted. Show the coefficient extraction for $\chi^{(3,1,1)}((2,2,1))$ explicitly, and check row orthogonality with weights $1/z_\mu$. The two S_4 entries already extracted in Lecture 13 need only be checked. In table headers, 31^2 and 2^21 denote $(3,1,1)$ and $(2,2,1)$.

Exercise 2 (Outer induction and multiplication of characteristics). For representations U of S_m and V of S_n, with $m, n \geq 1$, define their outer product character by

$$\chi_U \star \chi_V = \chi_{\mathrm{Ind}_{S_m \times S_n}^{S_{m+n}}(U \boxtimes V)}.$$

The external tensor product has action $(g,h)(u \otimes v) = gu \otimes hv$. Prove $\mathrm{ch}(\chi_U \star \chi_V) = \mathrm{ch}(\chi_U)\,\mathrm{ch}(\chi_V)$ in $N \geq m+n$ variables by counting invariant m-subsets of the cycles of a permutation. Compute $1_{S_2} \star \mathrm{sgn}$ using the S_4 table. Compare its group and degree with the inner tensor product $\mathbf{1} \otimes \mathrm{sgn}$ of representations of S_2.

Exercise 3 (Young permutation modules and Hall duality). For $\mu = (\mu_1, \ldots, \mu_t) \vdash n$, let $M^\mu = \mathrm{Ind}_{S_{\mu_1} \times \cdots \times S_{\mu_t}}^{S_n} \mathbf{1}$. Use the fixed ordered-block count to prove $\mathrm{ch}(\chi_{M^\mu}) = h_\mu$. Then prove $\langle h_\mu, m_\lambda \rangle = \delta_{\mu\lambda}$. For this second assertion expand the Cauchy kernel $\prod_{i,j}(1 - x_i y_j)^{-1}$ both in power sums and by monomials in y. Use enough variables for stability, as in Lecture 13.

Exercise 4 (Young's rule from Pieri). For diagrams $\nu \subseteq \lambda$, the *skew diagram* λ/ν is the set of boxes of λ outside ν. A horizontal strip is a skew diagram with at most one box in each column. Use as a given Pieri's rule

$$h_r s_\nu = \sum_{\lambda : \lambda/\nu \text{ horizontal of size } r} s_\lambda \qquad (r \geq 0),$$

with $s_\varnothing = h_0 = 1$ (Macdonald I.5, (5.16), p. 73). A semistandard tableau has weakly increasing rows and strictly increasing columns. Its weight μ counts entries of each label, and $K_{\lambda\mu}$ is the number of such tableaux of shape λ and weight μ. Prove Young's rule $M^\mu \cong \bigoplus_\lambda K_{\lambda\mu} V^\lambda$. Compute the decomposition for $\mu = (2,2,1)$ and check its dimension.

Exercise 5 (Removing border strips). A border strip is a non-empty connected skew diagram containing no 2×2 square, with connectivity along edges. Its height is the number of occupied rows minus one. For this exercise take as given the Murnaghan–Nakayama rule

$$\chi^\lambda((k,\nu)) = \sum_{\eta : \lambda/\eta \text{ border strip of size } k} (-1)^{\mathrm{ht}(\lambda/\eta)} \chi^\eta(\nu).$$

Here (k,ν) denotes cycle lengths together, rearranged into a partition. Compute the character value

$$\chi^{(3,2,1)}((3,3))$$

by listing all removable border strips of size three. Check the result with the Frobenius formula. Exercise 7 gives a separate proof of the rule.

Exercise 6 (The inner tensor product and Kronecker coefficients). For representations of the same S_n, define

$$g_{\lambda\mu\nu} = \langle \chi^{\nu}, \chi^{\lambda}\chi^{\mu} \rangle_{S_n}.$$

Explain why these are non-negative integers and why they give the multiplicities in $V^{\lambda} \otimes V^{\mu}$. Using the S_5 table, decompose $V^{(3,2)} \otimes V^{(3,2)}$. Explain why its characteristic is not the ordinary product $s^2_{(3,2)}$. The numbers above are called *Kronecker coefficients*; they concern the inner tensor product of representations, not outer induction.

Exercise 7 (⋆ Proving Murnaghan–Nakayama). Prove the rule of Exercise 5 from the alternant identity

$$p_k a_{\mu+\delta} = \sum_{i=1}^{N} a_{\mu+\delta+k\varepsilon_i}, \qquad k \geq 1,$$

where ε_i increments exponent i. Put $b_i = \mu_i + N - i$. Sort the new exponents after increasing one b_i by k. Show that a repeated exponent gives zero and that every other term is a border-strip addition λ/μ, with sign $(-1)^{\mathrm{ht}(\lambda/\mu)}$. Finish by comparing Schur coefficients in $p_k p_{\nu}$. This optional proof is independent of Exercise 8.

Exercise 8 (⋆ The hook-length formula). For $\lambda \vdash n \geq 1$, let $\lambda'_j = \#\{i : \lambda_i \geq j\}$ be the column lengths, the parts of the conjugate partition λ'. Let $h(i,j) = \lambda_i - j + \lambda'_j - i + 1$ be the hook length at box (i,j). Here $h(i,j)$ is a hook length; h_r and h_{μ} remain the complete symmetric functions. Derive from the Frobenius formula at cycle type (1^n) the determinant

$$f^{\lambda} = n! \det\left(\frac{1}{(\lambda_i - i + j)!} \right)_{i,j=1}^{N}, \qquad N = \ell(\lambda),$$

where a negative factorial has reciprocal zero. Evaluate it by putting $b_i = \lambda_i + N - i$ and using monic falling factorials. Prove the rowwise hook-set identity

$$\{h(i,j) : 1 \leq j \leq \lambda_i\} = \{1, \ldots, b_i\} \setminus \{b_i - b_t : i < t \leq N\}.$$

Deduce $f^{\lambda} = n! / \prod_{(i,j)\in\lambda} h(i,j)$ and compute $f^{(4,2,1)}$. This optional proof does not use Exercise 7.